

Solutions for Homework Six, CSE 355

1. (8.1, 10 points) Let M be the Turing machine defined by

δ	B	a	b	c
q_0	q_1, B, R			
q_1	q_2, B, L	q_1, a, R	q_1, c, R	q_1, c, R
q_2		q_2, c, L		q_2, b, L

- Trace the computation for the input string $aabca$.
- Trace the computation for the input string $bcbcb$.
- Give the state diagram of M .
- Describe the result of a computation in M .

Solution:

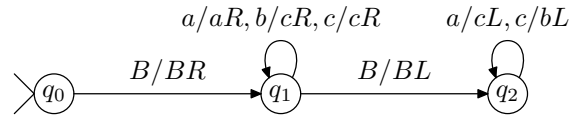
a)

$q_0BaabcaB$
 $\vdash Bq_1aabcaB$
 $\vdash Baq_1abcaB$
 $\vdash Baaq_1bcaB$
 $\vdash Baacq_1caB$
 $\vdash Baaccq_1aB$
 $\vdash Baaccaq_1B$
 $\vdash Baaccq_2aB$
 $\vdash Baacq_2ccB$
 $\vdash Baaq_2cbcB$
 $\vdash Baq_2abbcB$
 $\vdash Bq_2acbcbB$
 $\vdash q_2BccbcbB$

b)

$q_0BbcbcbB$
 $\vdash Bq_1bcbcbB$
 $\vdash Bcq_1cbcB$
 $\vdash Bccq_1bcB$
 $\vdash Bcccq_1cB$
 $\vdash Bccccq_1B$
 $\vdash Bcccq_2cB$
 $\vdash Bccq_2cbB$
 $\vdash Bcq_2cbbbB$
 $\vdash Bq_2cbbbbB$
 $\vdash q_2BbbbbbB$

- c) The state diagram of M is



- d) The result of a computation is to replace the a 's in the input string with c 's and the c 's with b 's.

■

2. (8.2, 10 points) Let M be the Turing machine defined by

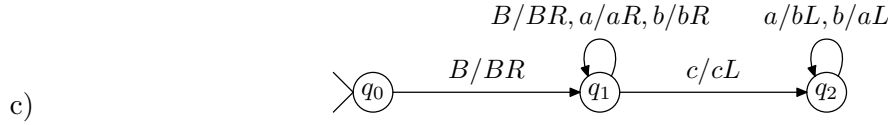
δ	B	a	b	c
q_0	q_1, B, R			
q_1	q_1, B, R	q_1, a, R	q_1, b, R	q_2, c, L
q_2		q_2, b, L	q_2, a, L	

- Trace the computation for the input string $abcb$.
- Trace the first six transitions of the computation for the input string $abab$.
- Give the state diagram of M .
- Describe the result of a computation in M .

Solution:

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$q_0 BabcbabB$	$q_0 BababB$
$\vdash Bq_1 abcbabB$	$\vdash Bq_1 ababB$
$\vdash Baq_1 bcbabB$	$\vdash Baq_1 babB$
$\vdash Babq_1 cabB$	$\vdash Babq_1 abB$
$\vdash Baq_2 bcbabB$	$\vdash Babaq_1 bB$
$\vdash Bq_2 aacabB$	$\vdash Bababq_1 B$
$\vdash q_2 BbacabB$	$\vdash BababBq_1$



- The result of a computation is to replace the a 's before the first c with b 's and the b 's before the first c with a 's. If there is no c in the input string, just keep the same.

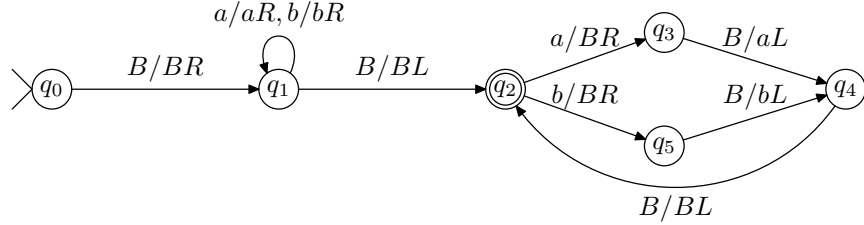
3. (8.3 (a, b, d), 20 points) Construct a Turing machine with input alphabet $\{a, b\}$ to perform each of the following operations. Note that the tape head is scanning position zero in state q_f whenever a computation terminates. ■

- Move the input one space to the right. Input configuration $q_0 BuB$, result $q_f BBuB$.
- Concatenate a copy of the reversed input string to the input. Input configuration $q_0 BuB$, result $q_f Buu^R B$.
- Erase the b 's from the input. Input configuration $q_0 BbabaababB$, result $q_f BaaaaB$.

Solution:

- a) Move the input one space to the right.

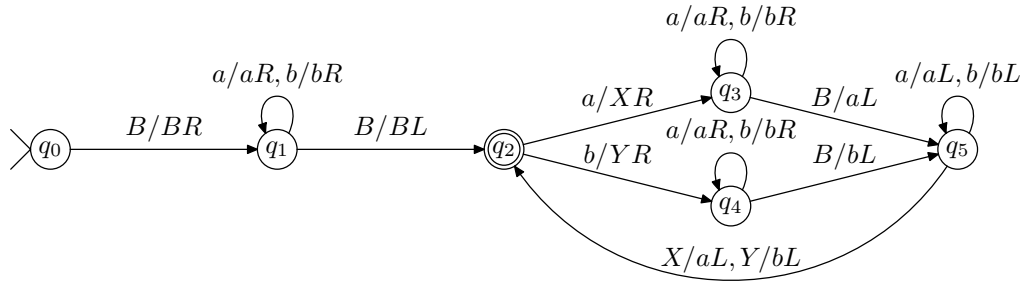
First the head moves to the first blank following the input by transitions to states q_0 and q_1 , and then moves to the left. At the state q_2 , starting with the rightmost symbol in the input and working in a right-to-left manner, the machine moves each symbol one position to the right.



If the input is the null string, the computation halts in the final state q_2 with the tape head in position zero as desired. Otherwise, an a is moved to the right by transitions to states q_3 and q_4 . Similarly, states q_5 and q_4 shift a b . This process is repeated until the entire string has been shifted.

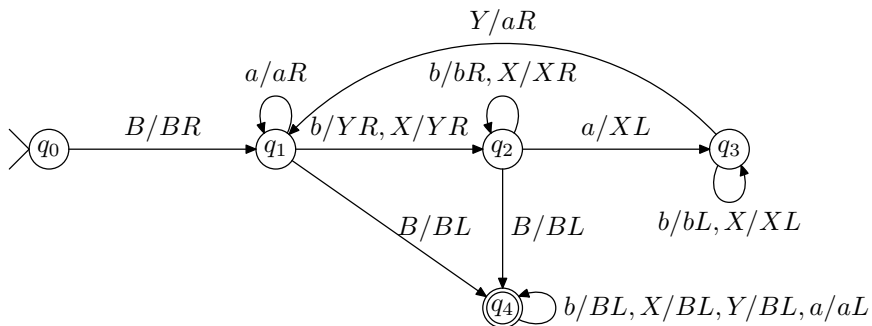
- b) Concatenate a copy of the reversed input string to the input.

First the head moves to the rightmost symbol in the input. At state q_2 , the machine works in a right-to-left manner, and copy a symbol to the end of the current string.



- d) Erase the b 's from the input.

The machine works in a left-to-right manner. At state q_1 , it skips a 's and whenever it reads a b or a previously deleted position, denoted by symbol X , the machine move the first a in the following string. If no further a is find, it goes to the final state q_4 and change all the remaining symbols b , X and Y into blanks.

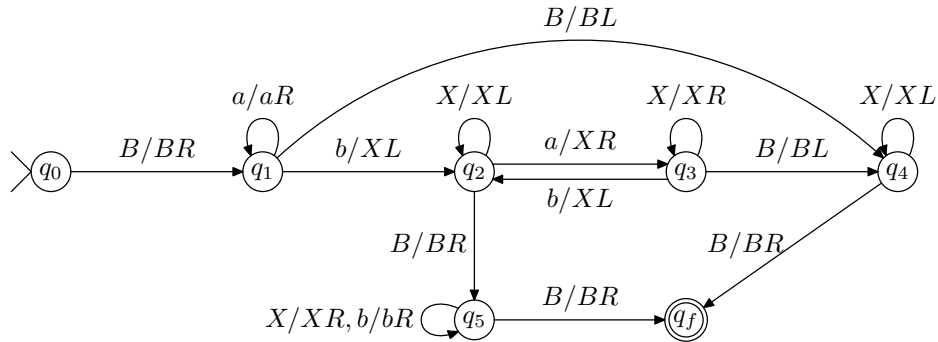


4. (8.5, 10 points) Construct a Turing machine with input alphabet $\{a, b\}$ to accept each of the following languages by final state.

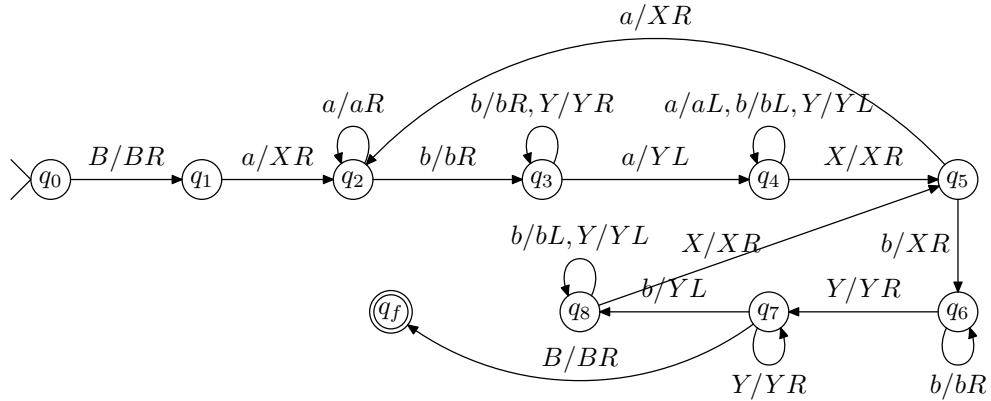
- a) $\{a^i b^j \mid i \geq 0, j \geq i\}$
- b) $\{a^i b^j a^i b^j \mid i, j > 0\}$
- c) Strings with the same number of a 's and b 's
- d) $\{uu^R \mid u \in \{a, b\}^*\}$
- e) $\{uu \mid u \in \{a, b\}^*\}$

Solution:

- a) $\{a^i b^j \mid i \geq 0, j \geq i\}$



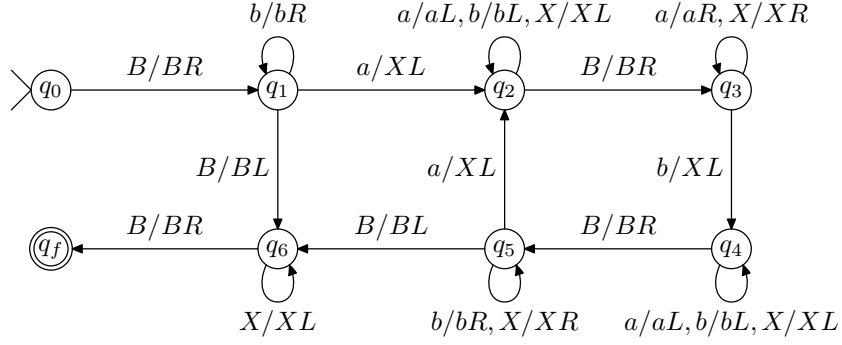
- b) $\{a^i b^j a^i b^j \mid i, j > 0\}$



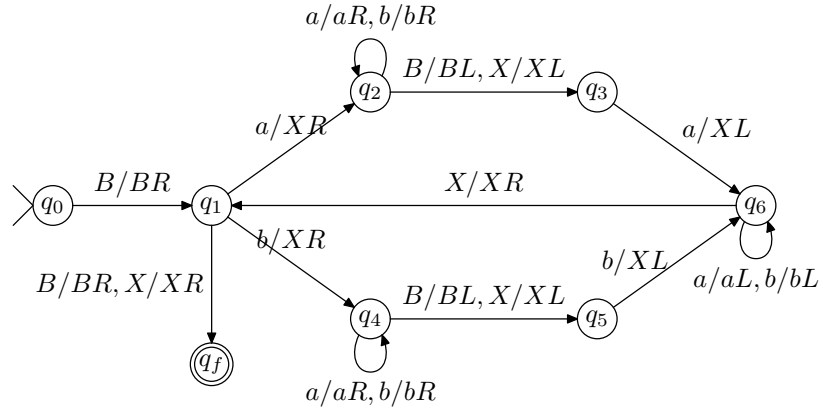
- c) Strings with the same number of a 's and b 's

A computation of the machine begins by finding the first a on the tape and replacing it with an X (state q_1). The tape head is then returned to position zero and a search is initiated for a corresponding b . If a b is encountered in state q_3 , an X is written and the tape head is repositioned to repeat the cycle q_2, q_3, q_4, q_5 . If no matching b is found, the computation halts in state q_3 rejecting the input. After all the a 's have been processed, the entire string

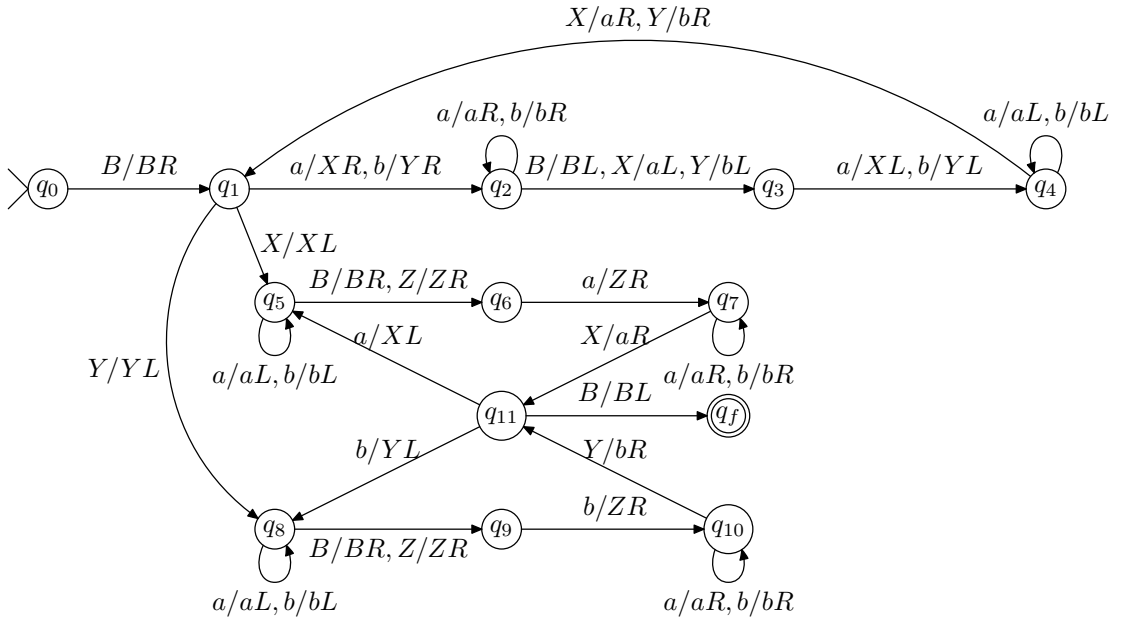
is read in q_5 and q_6 is entered upon reading the trailing blank. The computation halts in the accepting state q_f if no b 's remain on the tape.



d) $\{uu^R \mid u \in \{a, b\}^*\}$

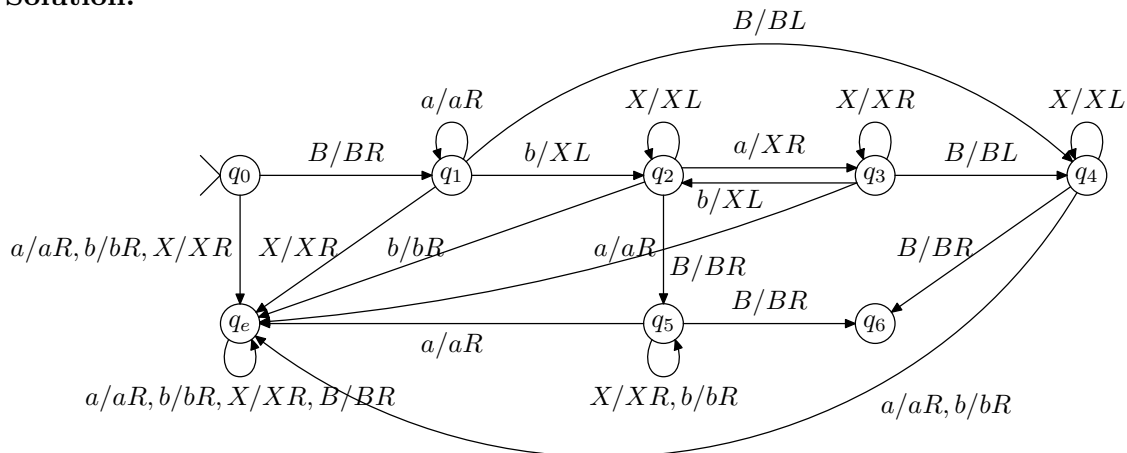


e) $\{uu \mid u \in \{a, b\}^*\}$



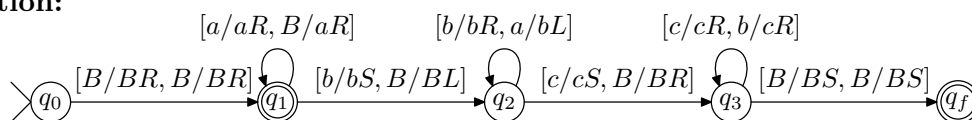
5. (8.6, 10 points) Modify your solution to Exercise 5(a) to obtain a Turing machine that accepts the language $\{a^i b^j \mid i \geq 0, j \geq i\}$ by halting.

Solution:



6. (8.16, 20 points) Construct a two-tape Turing machine with input alphabet $\{a, b, c\}$ that accepts the language $\{a^i b^i c^i \mid i \geq 0\}$.

Solution:



7. (8.18, 20 points) Construct a two-tape Turing machine that accepts strings in which each a is followed by an increasing number of b 's; that is, the strings are of the form

$$ab^{n_1} ab^{n_2} \dots ab^{n_k}, k > 0,$$

where $n_1 < n_2 < \dots < n_k$.

Solution:

